

Saber Ayatollahi, CFA, MSc

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SUMMARY

Associate Director, Enterprise Model Risk Management candidate with ~7 years across model validation, quantitative model development, and risk analytics. Currently operates within Moody's Analytics' formal model-governance framework with delegated sign-off on valuation, sensitivity, and risk outputs for multi-asset institutional portfolios. Hands-on Python and SQL practitioner; stochastic and Monte Carlo modelling foundations from dual MSc (Financial Modelling, Chemical Engineering); applied knowledge of OSFI E-23 and SR 11-7 validation standards. CFA charterholder.

CORE SKILLS

Model Validation & Independent Review • Model Risk & Governance (OSFI E-23 / SR 11-7) • Quantitative Risk & Statistical Modelling • Stochastic Modelling & Monte Carlo Simulation • Benchmark Model Development • Python (pandas, NumPy, SciPy) • SQL & Large-Dataset Analytics • Model Documentation & Governance Reporting • Stress Testing & Scenario Analysis • Git / Version Control • VaR & CVaR Analytics

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services | Toronto, May 2023 - Present

Independent Model Validation & Challenge

- Holds delegated sign-off authority on valuation, sensitivity, and risk outputs for multi-asset institutional client portfolios ranging \$5-25bn per engagement; independently reviews curve construction, spread calibration, and cross-asset interactions prior to production release.
- Operates within a formal model-risk and governance framework – challenges assumptions, assesses economic and mathematical defensibility, and escalates outputs lacking quantitative support; maintains documentation standards consistent with SR 11-7 and OSFI E-23 expectations.
- Validates derivatives pricing outputs across rates, FX, and inflation; cross-checks sensitivities and scenario impacts against benchmark calculations at portfolio-level aggregates.

Quantitative Model Development & Benchmark Analysis

- Led design and implementation of a multi-scenario cash-flow projection engine supporting base, stress, and reverse-stress scenarios – embedded behavioural assumptions, prepayment logic, and macro stress overlays.
- Re-engineered manual spreadsheet workflows into auditable Python analytics pipelines under Git version control, with embedded logging, validation tests, and reconciliation against shadow runs to satisfy model-governance audit.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection – reducing validation cycle time by an estimated 30-40% on comparable modules.

Validation Reporting & Stakeholder Recommendations

- Prepared comprehensive model documentation and validation summaries for senior stakeholders covering scenario impacts, sensitivities, and methodology observations; serves as escalation point on client-critical analytical issues, interfacing with Heads of Risk and investment stakeholders.

Client Service Specialist | Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform; validated model outputs post-deployment.
- Translated client investment-team requirements into structured Product Owner requests; advised on asset, liability, and risk-model configurations across the assigned book of accounts.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation | Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients – aligned regulatory requirements with system implementation and contributed to governance documentation and regulatory-readiness processes.
- Coordinated across actuarial, finance, IT, and risk teams on milestone delivery, implementation materials, and control documentation.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk | Toronto, Feb 2019 - Sep 2021

- Built and interpreted stochastic economic scenario generators using Monte Carlo techniques and statistical calibration to assess interest-rate, inflation, and currency risk under base-case and stressed environments.
- Performed VaR and CVaR portfolio optimization (asset-only and asset-liability surplus) on the GLASS platform; conducted risk decomposition, contribution-to-risk, and near-optimal-frontier robustness analysis for institutional clients.
- Delivered ALM studies for Canadian and international pension funds – analyzed funding ratios, duration mismatch, and long-term balance-sheet risk exposures; presented findings to investment committees on-site.
- Led modelling and validation work for a three-plan university pension merger (JSPP) on a team of three – captured all Funding Policy dynamics and tested duration, currency, inflation, and leverage overlay strategies.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016